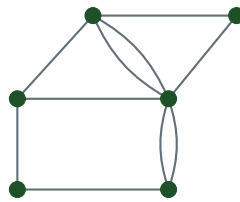


CS-5136: Graph Theory

Operations on Graphs

pages 51–60 · union, intersection, ring sum & Cartesian product — with detailed step-by-step proofs and a fully self-contained side trip into matching theory



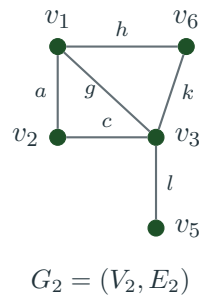
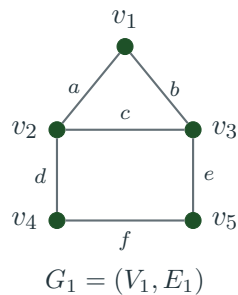
Built from: A. Bhattacharjee, “CS-5136: Graph Theory”, NIT Silchar
matching theory (Section 4) is built from scratch here — no other book required (that field is traditionally covered via
S. Held, “Combinatorial Optimization”, Uni Bonn)

1 Why Operations on Graphs?

Every kind of mathematical object comes with operations for combining or transforming it into new objects of the same kind: numbers can be added, functions composed, sets intersected. Graphs are no different — given two graphs, we'd like standard ways to combine them into a new graph. Today covers the four such operations from your slides: **union**, **intersection**, **ring sum**, and **Cartesian product**.

Running example

Every operation below is applied to the *same* pair of graphs G_1, G_2 , so you can watch how each one changes. G_1 and G_2 deliberately share some vertices (v_1, v_2, v_3, v_5) and even some *identically-named* edges (a, c) — that overlap is exactly what makes the four operations produce different results from each other.



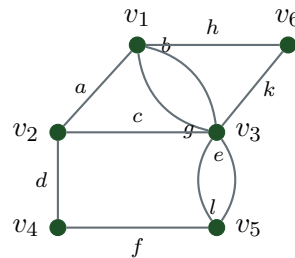
2 The Four Operations

2.1 Union

Definition 2.1 (Union)

The **union** of $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is the graph $G_3 = G_1 \cup G_2$ with $V_3 = V_1 \cup V_2$ and $E_3 = E_1 \cup E_2$.

EXAMPLE — $G_3 = G_1 \cup G_2$



$V_3 = \{v_1, \dots, v_6\}$ (6 vertices: the shared v_1, v_2, v_3, v_5 plus G_1 's private v_4 and G_2 's private v_6). E_3 has 10 edges: a and c are the *same* edge in both graphs, so they appear once each; but b (from G_1) and g (from G_2) are *different* edges that happen to join the same pair v_1, v_3 — so in the union they survive as a pair of **parallel edges**. Likewise e and l become parallel edges between v_3, v_5 .

2.2 Intersection

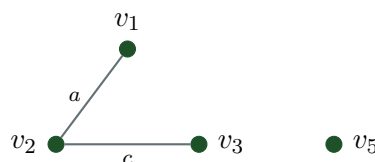
Definition 2.2 (Intersection)

The **intersection** $G_1 \cap G_2$ is the graph G_4 consisting only of the vertices and edges that belong to *both* G_1 and G_2 .

The subtlety worth catching

“Belongs to both” is about edge *identity* (the label), not just endpoints. $e = (v_3, v_5)$ is in G_1 and $l = (v_3, v_5)$ is in G_2 — same endpoints, but they are *different* edges, so *neither* survives into the intersection. Only a and c — literally the same edge in both graphs — do.

EXAMPLE — $G_4 = G_1 \cap G_2$



$V_4 = \{v_1, v_2, v_3, v_5\}$ (every vertex common to both — v_5 is included since $v_5 \in V_1 \cap V_2$, even though it ends up **isolated**, since neither of its edges survived). $E_4 = \{a, c\}$.

2.3 Ring sum

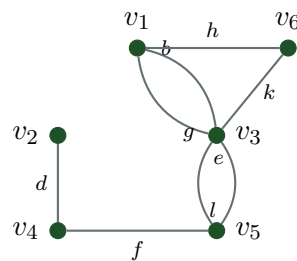
Definition 2.3 (Ring sum)

The **ring sum** $G_1 \oplus G_2$ is the graph G_5 with $V = V_1 \cup V_2$ and edge set equal to those edges in G_1 or G_2 , **but not both**.

A preview

$E(G_1 \oplus G_2) = (E_1 \setminus E_2) \cup (E_2 \setminus E_1)$ — exactly the **symmetric difference** of the two edge sets. This same operation, written $\triangle$, is the single most useful tool in *matching theory* (a branch of combinatorial optimization). Section 4 below builds that whole topic from scratch — definitions, examples, and a full proof — purely so that today’s practice questions (Set C) are self-contained and don’t send you hunting for another book.

EXAMPLE — $G_5 = G_1 \oplus G_2$



Same vertex set as the union (6 vertices), but a and c — the edges common to both — have vanished, leaving 8 edges: $\{b, d, e, f, g, h, k, l\}$. Vertex v_2 keeps only d ; it lost a .

2.4 Cartesian product

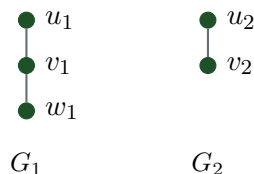
Definition 2.4 (Cartesian product)

The **Cartesian product**^a $G_1 \boxtimes G_2$ is the graph G with $V = V_1 \times V_2$, where distinct vertices (u, v) and (x, y) are adjacent exactly when **either** (1) $u = x$ and $vy \in E(G_2)$, **or** (2) $v = y$ and $ux \in E(G_1)$.

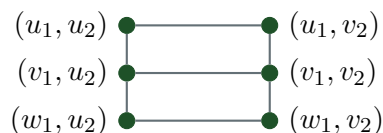
^aMost textbooks write this as $G_1 \square G_2$; your slides use $G_1 \boxtimes G_2$ for the same thing — just be aware $\boxtimes$ means something different (the *strong* product) in most other sources.

EXAMPLE — $P_3 \boxtimes P_2$

Let G_1 be the path $u_1 - v_1 - w_1$ and G_2 the path $u_2 - v_2$.



The product has $3 \times 2 = 6$ vertices, one per pair, forming a 3×2 grid (a “ladder”):



Horizontal edges come from rule (1) (same G_1 -coordinate, an edge in G_2); vertical edges from rule (2) (same G_2 -coordinate, an edge in G_1). Counting: 3 horizontal + 4 vertical = 7 edges — matches the formula proved in Theorem 3.3 below.

3 Provable Properties

None of this is stated as a theorem in your slides, but every one of these is a natural (and examinable) fact about the four operations. Below, every proof is broken into small numbered steps — nothing is skipped. $n_i = |V_i|$, $m_i = |E_i|$ throughout.

Theorem 3.1 (Counting for union and intersection)

$$|V(G_1 \cup G_2)| = n_1 + n_2 - |V_1 \cap V_2|, \quad \|G_1 \cup G_2\| = m_1 + m_2 - \|G_1 \cap G_2\|.$$

Proof. We first prove a general fact about counting, then apply it twice.

The general fact. Suppose A and B are two finite collections of things (could be vertices, could be edges, could be anything), and some things happen to be in *both* A and B . We want a formula for $|A \cup B|$, the number of distinct things in A together with B .

1. Start by just adding $|A| + |B|$.
2. Every item that is *only* in A (not in B) gets counted once in this sum — correct.
3. Every item that is *only* in B gets counted once too — also correct.
4. But every item that is in *both* A and B gets counted **twice**: once when we added $|A|$, and once more when we added $|B|$. It should only be counted *once* in $A \cup B$.
5. There are exactly $|A \cap B|$ such doubly-counted items. So our sum $|A| + |B|$ is too big by exactly $|A \cap B|$.
6. Subtracting that excess gives the correct count:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Applying it. This fact doesn't care what kind of things A and B contain — it works for any finite collection. So:

- Let $A = V_1$ and $B = V_2$ (the vertex sets). The fact says $|V_1 \cup V_2| = |V_1| + |V_2| - |V_1 \cap V_2| = n_1 + n_2 - |V_1 \cap V_2|$. Since $V(G_1 \cup G_2) = V_1 \cup V_2$ by Definition 2.1, this proves the first identity.
- Let $A = E_1$ and $B = E_2$ (the edge sets). The same fact gives $|E_1 \cup E_2| = m_1 + m_2 - |E_1 \cap E_2|$. Since $E(G_1 \cup G_2) = E_1 \cup E_2$ and $\|G_1 \cap G_2\| = |E_1 \cap E_2|$ (Definition 2.2), this proves the second identity. ■

EXAMPLE

Let's check the formula catches no one by surprise, using G_1, G_2 from Section 1. $n_1 = n_2 = 5$. $V_1 \cap V_2 = \{v_1, v_2, v_3, v_5\}$ (every vertex appearing in both pictures), so $|V_1 \cap V_2| = 4$. Formula: $|V_3| = 5 + 5 - 4 = 6$. Direct count from the union picture (Section 2.1): $\{v_1, \dots, v_6\}$, that's 6. ✓

Now edges: $m_1 = m_2 = 6$. $E_1 \cap E_2 = \{a, c\}$ (the only two edges with the exact same name in both lists), so $\|G_1 \cap G_2\| = 2$. Formula: $\|G_3\| = 6 + 6 - 2 = 10$. Direct count from Section 2.1: $\{a, b, c, d, e, f, g, h, k, l\}$, that's 10. ✓

Theorem 3.2 (Counting for ring sum)

$$\|G_1 \oplus G_2\| = m_1 + m_2 - 2\|G_1 \cap G_2\|.$$

Proof.

1. By Definition 2.3, an edge belongs to $G_1 \oplus G_2$ exactly when it is in E_1 or E_2 , *but not in both*. In other words: start from every edge that is in *at least one* of the two graphs, i.e. $E_1 \cup E_2$, and then *remove* every edge that is in both.
2. From Theorem 3.1, the size of that starting pile is $|E_1 \cup E_2| = m_1 + m_2 - \|G_1 \cap G_2\|$.
3. Now remove the edges that are in both graphs. There are $\|G_1 \cap G_2\|$ of them. Removing them means subtracting $\|G_1 \cap G_2\|$ *one more time* from what we had in Step 2 (this is a genuine removal — these edges are not wanted in the ring sum at all, unlike in the union where they were merely *not double-counted*).
4. Putting Steps 2 and 3 together:

$$\|G_1 \oplus G_2\| = (m_1 + m_2 - \|G_1 \cap G_2\|) - \|G_1 \cap G_2\| = m_1 + m_2 - 2\|G_1 \cap G_2\|.$$

■

EXAMPLE

Same numbers as before: $m_1 = m_2 = 6$, $\|G_1 \cap G_2\| = 2$. Formula: $\|G_1 \oplus G_2\| = 6 + 6 - 2(2) = 8$. Direct count from Section 2.3: $\{b, d, e, f, g, h, k, l\}$, that's 8. ✓

Theorem 3.3 (Counting for Cartesian product)

$$|V(G_1 \boxtimes G_2)| = n_1 n_2, \quad \|G_1 \boxtimes G_2\| = n_1 m_2 + n_2 m_1.$$

Proof. Vertices. By Definition 2.4, $V = V_1 \times V_2$, the set of *all* pairs (u, v) with $u \in V_1$, $v \in V_2$. There are n_1 choices for u and, independently, n_2 choices for v , so by the basic counting principle (choices multiply) there are $n_1 \times n_2$ such pairs in total.

Edges. There are two separate rules in Definition 2.4 for when an edge exists. We count how many edges each rule creates, one at a time.

1. *Rule (1) edges.* This rule connects (u, v) to (u, y) whenever $vy \in E(G_2)$ — notice the first coordinate u never changes. So fix one particular vertex $u \in V_1$ and look only at the vertices of the form $(u, *)$: there are exactly n_2 of them, one per vertex of G_2 , and Rule (1) joins two of them exactly when their second coordinates are joined in G_2 . So this one frozen slice reproduces G_2 exactly, edge for edge: it contributes m_2 Rule-(1) edges.
2. There are n_1 different vertices u we could have frozen, and each gives its own separate slice with its own m_2 edges (a different u means a completely different, non-overlapping set of vertices $(u, *)$). So the total number of Rule-(1) edges, across every choice of u , is $n_1 \times m_2$.
3. *Rule (2) edges.* By the mirror-image argument — now freezing a vertex $v \in V_2$ and varying the first coordinate according to $E(G_1)$ — there are $n_2 \times m_1$ Rule-(2) edges.
4. *Do the two rules ever create the same edge twice?* A Rule-(1) edge changes only the second coordinate (first coordinate frozen). A Rule-(2) edge changes only the first coordinate (second coordinate frozen). For one edge to be *both* at once, it would have to change neither coordinate — but an edge always joins

two *different* vertices, so something must change. Hence no edge is ever counted by both rules: the two piles (Rule-1 edges and Rule-2 edges) are completely separate.

5. Since the two piles don't overlap, the total edge count is simply their sum:

$$\|G_1 \boxtimes G_2\| = n_1 m_2 + n_2 m_1.$$

■

EXAMPLE

From Section 2.4: $n_1 = 3$ (vertices u_1, v_1, w_1), $m_1 = 2$ (two edges in the path); $n_2 = 2$ (vertices u_2, v_2), $m_2 = 1$ (one edge). Formula: vertices = $3 \times 2 = 6$; edges = $n_1 m_2 + n_2 m_1 = 3(1) + 2(2) = 3 + 4 = 7$. Direct count from the ladder picture: 3 horizontal + 4 vertical = 7. ✓

Two quick remarks

Union, intersection, and ring sum are all built directly from *set* union/intersection/symmetric-difference on V_i and E_i — so they instantly inherit commutativity ($G_1 \cup G_2 = G_2 \cup G_1$, etc.) from the corresponding set identities. The Cartesian product is commutative too, but only up to *isomorphism*: swapping coordinates, $(u, v) \mapsto (v, u)$, is a bijection $V_1 \times V_2 \rightarrow V_2 \times V_1$ that preserves adjacency, so $G_1 \boxtimes G_2 \cong G_2 \boxtimes G_1$.

4 Background: Matchings and the $\triangle$ Operation

Practice Set C (end of this document) connects ring sum to a topic called *matching theory*. That topic isn't part of your CS-5136 syllabus — it belongs to a different field (combinatorial optimization) — so rather than send you elsewhere for it, this section defines everything about it from scratch, with a full proof. Once you've read this section, Set C needs nothing outside this PDF.

4.1 Definitions

Definition 4.1 (Matching)

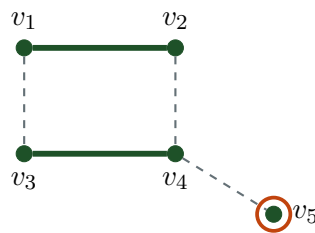
Given a graph $G = (V, E)$, a **matching** M is a set of edges of G such that no two edges of M share an endpoint — equivalently, every vertex touches *at most one* edge of M .

Definition 4.2 (M -covered / M -exposed)

A vertex touched by some edge of M is **M -covered**; a vertex touched by no edge of M is **M -exposed**.

EXAMPLE

Let H be the graph with $V = \{v_1, \dots, v_5\}$ and edges $(v_1, v_2), (v_1, v_3), (v_2, v_4), (v_3, v_4), (v_4, v_5)$:



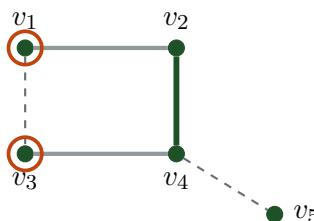
$M = \{(v_1, v_2), (v_3, v_4)\}$ (solid, green) is a matching: check each vertex touches at most one solid edge — yes. v_1, v_2, v_3, v_4 are M -covered; v_5 (ringed) is M -exposed.

Definition 4.3 (Alternating / augmenting path)

Given a matching M , a path P is **M -alternating** if its edges go in- M , not-in- M , in- M , ..., alternating all the way along. P is **M -augmenting** if, in addition, *both* of its endpoints are M -exposed.

EXAMPLE

Same graph H , now with the smaller matching $M' = \{(v_2, v_4)\}$ and the path $P = v_1 - v_2 - v_4 - v_3$:



P 's edges are (v_1, v_2) [not in M'], (v_2, v_4) [in M' , bold green], (v_4, v_3) [not in M'] — alternating. Both endpoints v_1, v_3 (ringed) are M' -exposed. So P is M' -augmenting.

4.2 The key lemma

Lemma 4.4

If M is a matching and P is an M -augmenting path, then $M \triangle P$ (flip every edge of P : drop its M -edges, add its non- M -edges, leave everything else in M untouched) is also a matching, and $|M \triangle P| = |M| + 1$.

Proof. Step 1 — what does the flip do? $M \triangle P$ keeps every edge of M that is not on P , removes every edge of M that is on P , and adds every edge of P that was not already in M .

Step 2 — checking every vertex still touches at most one edge (i.e., it really is a matching). We check three kinds of vertex separately.

- *Vertices not on P at all.* Nothing about them changed — they touch exactly the same M -edges as before. Since M was a matching, they're still fine.
- *Internal vertices of P* (every vertex of P except its two ends). Since P is a path, such a vertex touches exactly *two* of P 's edges: the one before it and the one after it in the walk. Because P is alternating, exactly one of those two is an M -edge and the other is not. After flipping, the M -edge disappears and the non- M -edge appears — so this vertex still touches exactly *one* edge afterward, just a different one. Still fine.
- *The two endpoints of P .* Each endpoint is M -exposed (that's what "augmenting" means), so it touched *zero* M -edges before. Each endpoint touches exactly *one* edge of P (the first or last edge of the walk). Since both endpoints started exposed, that first/last edge cannot be an M -edge (an M -edge there would mean the endpoint was already covered) — so it's a non- M -edge, and it gets *added*. Each endpoint goes from touching 0 edges to touching exactly 1. Still fine.

Every vertex touches at most one edge of $M \triangle P$ in every case — so $M \triangle P$ is a matching.

Step 3 — counting the size. Say P has ℓ edges. Because P alternates and *starts and ends* on a non- M -edge (Step 2's argument for the endpoints), the non- M -edges outnumber the M -edges on P by exactly one: there are $\frac{\ell+1}{2}$ non- M -edges and $\frac{\ell-1}{2}$ M -edges. Flipping removes the $\frac{\ell-1}{2}$ M -edges of P from M and adds the $\frac{\ell+1}{2}$ non- M -edges. Net change in size:

$$-\frac{\ell-1}{2} + \frac{\ell+1}{2} = \frac{(\ell+1) - (\ell-1)}{2} = \frac{2}{2} = 1.$$

So $|M \triangle P| = |M| + 1$. ■

EXAMPLE — Checking Lemma 4.4 on the example above

$M' = \{(v_2, v_4)\}$, $|M'| = 1$. $P = v_1 v_2 v_4 v_3$ has $\ell = 3$ edges: 1 M' -edge ($\frac{3-1}{2} = 1$) and 2 non- M' -edges ($\frac{3+1}{2} = 2$). Flip: drop (v_2, v_4) , add (v_1, v_2) and (v_3, v_4) . Result: $M' \triangle P = \{(v_1, v_2), (v_3, v_4)\}$, size $2 = 1 + 1$. ✓ (And notice: this is exactly the matching M from the very first example above!)

4.3 Why this matters: Berge's theorem

Theorem 4.5 (Berge's Theorem --- statement only)

A matching M of G has the largest possible size (among all matchings of G) if and only if G has no M -augmenting path.

The full proof of Theorem 4.5 is longer than we need here — but one specific building-block inside that proof is exactly what Practice Question 7 asks you to explore, so we prove *that* piece in full:

Lemma 4.6

If M and M' are both matchings of the same graph G , then $H = M \triangle M'$ (as a set of edges) splits into pieces that are each either a simple path or an *even*-length cycle, with edges alternating between M and M' along each piece.

Proof. Step 1. Take any vertex v . At most one edge of M touches v (since M is a matching), and at most one edge of M' touches v (since M' is a matching too). If v happens to touch the *same* edge in both M and M' , that edge cancels out of the symmetric difference entirely (it's in *both*, so it is not in $M \triangle M'$). So v touches **at most 2** edges of H .

Step 2. A graph in which every vertex has degree at most 2 must be a disjoint union of simple paths and cycles. (Why: start walking from any vertex along an unused edge; at every vertex you arrive at, there is at most one *other* unused edge to leave by, since degree ≤ 2 means at most 2 edges touch it total, and you just used one of them to arrive. So the walk can't branch. It must eventually either run out of edges — ending the walk at a path's endpoint — or return to where it started, closing a cycle.)

Step 3. Along any one such path or cycle, consecutive edges must alternate between M and M' : if two consecutive edges sharing a vertex were *both* from M , that vertex would touch two M -edges, contradicting that M is a matching (symmetrically for M'). So no two consecutive H -edges can be the same type — they alternate.

Step 4. A cycle's edges alternate all the way around and must return to the same edge-type they started with (to close up consistently) — which forces the total number of edges in any such cycle to be *even*. ■

5 Day 3 Cheat-Sheet

Recap

- Union: $V_1 \cup V_2, E_1 \cup E_2$. Same-endpoint edges with *different* names survive as parallel edges.
- Intersection: only vertices/edges in *both* — edge identity matters, not just endpoints.
- Ring sum: symmetric difference of edge sets — the same $\triangle$ as in matching theory (Section 4).
- Cartesian product: $V_1 \times V_2$; adjacency = “move along one coordinate the way the corresponding graph allows.”
- Counting: $|V_1 \cup V_2| = n_1 + n_2 - |V_1 \cap V_2|$; $\|G_1 \oplus G_2\| = m_1 + m_2 - 2\|G_1 \cap G_2\|$; $\|G_1 \boxtimes G_2\| = n_1 m_2 + n_2 m_1$.
- Matching M : pairwise disjoint edges. $M \triangle P$ for an augmenting path P is a matching of size $|M| + 1$ (Lemma 4.4). $M \triangle M'$ for two matchings splits into paths and *even* cycles (Lemma 4.6).

6 Practice Questions

Set A --- straight from today's definitions

Q1. Using G_1, G_2 from Section 1: is d (the edge (v_2, v_4)) present in $G_1 \cap G_2$? Why or why not?

A. No. $v_4 \notin V_2$ at all, so d can't possibly be an edge of G_2 , let alone a shared one.

Q2. Compute $d(v_3)$ (the degree of v_3) in G_1 , in G_2 , and in $G_1 \cup G_2$. Is it just the sum of the first two?

A. In G_1 : $d(v_3) = 3$ (b, c, e). In G_2 : $d(v_3) = 4$ (g, k, c, l). Naively summing gives 7, but the union only has degree 6 ($\{b, c, e, g, k, l\}$) — one less, because c is the same edge in both graphs and got counted twice in $3 + 4$. Moral: degrees add directly only when no edge at that vertex is shared between the two graphs; otherwise subtract one for each shared edge at that vertex.

Q3. Draw $P_2 \boxtimes P_2$ (two copies of a single edge). How many vertices and edges does it have, and what shape is it?

A. $n_1 = n_2 = 2, m_1 = m_2 = 1$. Vertices: $2 \times 2 = 4$. Edges: $n_1 m_2 + n_2 m_1 = 2(1) + 2(1) = 4$. Four vertices, four edges, each vertex degree 2 — that's the cycle graph C_4 (a square).

Set B --- properties

Q4. If G_1 and G_2 have completely disjoint vertex sets (no shared vertices at all, so certainly no shared edges), what do $G_1 \cap G_2$, $G_1 \cup G_2$, and $G_1 \oplus G_2$ look like?

A. $G_1 \cap G_2$ is the null graph (no vertices, no edges) — or depending on convention, simply undefined/empty. $G_1 \cup G_2$ and $G_1 \oplus G_2$ coincide exactly (since $E_1 \cap E_2 = \emptyset$ means nothing gets removed): both equal the graph with $V_1 \cup V_2$ and $E_1 \cup E_2$, i.e. G_1 and G_2 drawn side by side with no interaction.

Q5. Using Theorem 3.1: if $\|G_1 \cup G_2\| = 9, m_1 = 6, m_2 = 5$, what is $\|G_1 \cap G_2\|$?

A. $9 = 6 + 5 - \|G_1 \cap G_2\| \Rightarrow \|G_1 \cap G_2\| = 11 - 9 = 2$.

Set C --- matching theory (uses Section 4 above)

Q6. Using Theorem 3.2's counting formula (with $A = M$, $B = E(P)$ playing the role of E_1, E_2): if $|M| = 4$, a path P has 5 edges, and M, P share 2 edges in common, how many edges does $M \triangle P$ have? Does this match what Lemma 4.4 predicts if P happens to be M -augmenting (recall: $|M \triangle P| = |M| + 1$ in that case)?

A. $|M \triangle P| = |M| + |E(P)| - 2|M \cap E(P)| = 4 + 5 - 2(2) = 5$. For P to be M -augmenting it must alternate with one more non- M edge than M -edges (Lemma 4.4's proof, Step 3): with 5 edges total and 2 shared with M , that's 3 non- M edges and 2 M -edges — exactly that pattern. And indeed $5 = 4 + 1 = |M| + 1$. ✓ Same formula (Theorem 3.2) used in a completely different context, same correct answer.

Q7. Recall the graph H from Section 4.1 ($V = \{v_1, \dots, v_5\}$, edges (v_1, v_2) , (v_1, v_3) , (v_2, v_4) , (v_3, v_4) , (v_4, v_5)). Let $M = \{(v_1, v_2), (v_3, v_4)\}$ and $M' = \{(v_1, v_3), (v_2, v_4)\}$ (two different matchings of H , each of size 2). Compute $M \cup M'$, $M \cap M'$, and $M \triangle M'$ as edge sets, check Theorem 3.2's count against direct listing, then verify Lemma 4.6's claim (paths and even cycles) on this specific example.

A. $M \cap M' = \emptyset$ (no edge appears in both lists). $M \cup M' = \{(v_1, v_2), (v_3, v_4), (v_1, v_3), (v_2, v_4)\}$, size 4. $M \triangle M'$: since the intersection is empty, nothing gets removed, so $M \triangle M' = M \cup M'$ too — all 4 edges. Theorem 3.2 check: $|M| + |M'| - 2|M \cap M'| = 2 + 2 - 2(0) = 4$. ✓

For Lemma 4.6: list the 4 edges of $H' = M \triangle M'$ and follow them around — (v_1, v_2) , then (v_2, v_4) , then (v_4, v_3) [same edge as (v_3, v_4)], then (v_3, v_1) [same edge as (v_1, v_3)], back to v_1 . That's a single closed loop $v_1 \rightarrow v_2 \rightarrow v_4 \rightarrow v_3 \rightarrow v_1$ using all 4 edges — a 4-cycle, which is indeed even, exactly as Lemma 4.6 guarantees. Check the alternation too: $(v_1, v_2) \in M$, $(v_2, v_4) \in M'$, $(v_3, v_4) \in M$, $(v_1, v_3) \in M'$ — M, M', M, M' , alternating all the way round. ✓

Just say “Day 4” whenever you're ready for the next set (pages 61 onward, Algebraic treatment: isomorphism and beyond).